PHYS 101&103
Spring 2013-2014

Final Exam

PHYSICS 101 NO: _____

NAME: _____

(On my honor I have neither given nor received any unauthorized assistance on this exam.)

- This exam is out of 140 points which means that you have 40 bonus points.
- This exam is for 120 minutes and has two questions and four problems (Check this now carefully.)
- This is a closed-book/notes/calculator/laptop exam.
- Read the questions carefully and write down your solutions in the space provided neatly and clearly. Write down your final results in the boxes at the end of each question.
- Answers without work shown will not be given any credit.
- A formula sheet is provided for you.
- **SWITCH YOUR MOBILE PHONES OFF!!!** and put them on your desks.

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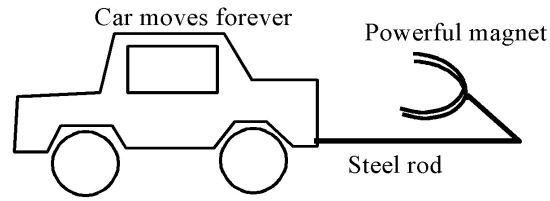
Question 1	Question 2	Problem 1	Problem 2	Problem 3	Problem 4	Total

Conceptual Question 1 (10 points)



In the figure, it says "car moves forever" due to the magnetic force? Please explain the conceptual mistake here.

If the magnet attracts the car, the car also attracts the magnet. However, the steel rod that connects the car and the magnet does not let them to come together. If we look at the forces on the car;

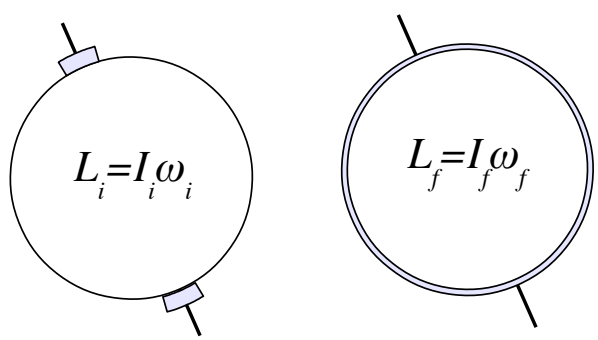
1.) Magnetic force (to the right)
2.) Force exerted by the rod (to the left and equal to the magnetic force in magnitude)

Hence, the car does not move forever due to the magnetic force!

Conceptual Question 2 (10 points)

A severe rise of global temperatures would eventually melt the Earth's polar ice caps. According to this prospective fact, days are being expected to be longer in the future. Considering Earth's rotation and associated conservation law, explain this phenomenon.

If the polar ice caps melt, the moment of inertia (rotational inertia) of Earth will be greater (see figure).



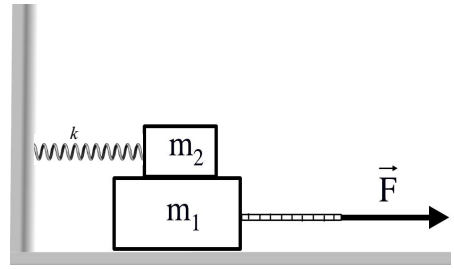
$$L_i = L_f \qquad I_i < I_f \quad \Longrightarrow \quad \omega_i > \omega_f$$

Days are longer

Angular momentum is conserved, because there is no net external torque on Earth with respect to its axis of rotation.

Problem (1) (30 points)

A block of mass $m_2 = 2.0$ kg which is tied to a wall with a light spring of force constant k = 1000.0 N/m, is sitting on another block with mass $m_1 = 10.0$ kg. An external force of $F = 45.0$ N was suddenly applied to the lower block when both masses were at rest and the spring was neither compressed nor stretched.

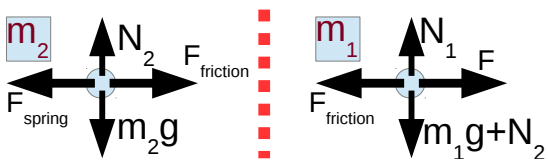


The block of mass m_2 moves to the right for 1.0 cm without slipping on the block of mass m_1. Then, it suddenly starts to slip on top of mass m_1 with an initial acceleration of 2.5 m/s^2 to the left. *(All measurements are made with respect to ground)*

The coefficients of kinetic and static friction between the masses are μ_k and μ_s respectively. There is **no** friction between m_1 and the floor. ($g \simeq 10.0$ m/s^2)

(a) (15 pts.) Find the coefficients of friction, kinetic and static, i.e, calculate μ_k and μ_s.

The only friction force is due to the surface between m_2 and m_1. Free Body diagrams for blocks of masses m_2 and m_1:



While moving together, the friction force is due to static friction. Thus $F_f \leq m_2 g \mu_s$. At the moment when m_2 starts slipping on m_1, its velocity and acceleration should be zero and the static friction force is at its maximum value. In this case $F_{spring} + F_{static\ friction} = 0 \rightarrow -kx + m_2 g \mu_s = 0 \rightarrow \boxed{\mu_s = kx/m_2 g = 0.5}$

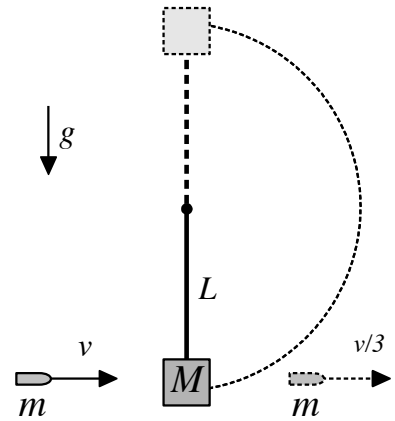
When static friction force cannot compansate the spring force, m_2 starts slipping on m_1. While surfaces are moving with respect to each other, kinetic friction force is present. At this moment the inital acceleration is known ($a_2 = -2.5\ m/s^2$). So: $F_{spring} + F_{kinetic\ friction} = m_2 a \rightarrow -kx + m_2 g \mu_k = ma \rightarrow \boxed{\mu_k = (kx + ma)/m_2 g = 0.25}$

(b) (15 pts.) Calculate the acceleration of block with mass m_1, just immediately after m_2 starts sliding.

On m_1 the friction force acts in opposite direction. Thus $F - F_{static\ friction} = m_1 a \rightarrow \boxed{a_{m_1} = (F - m_2 g \mu_k)/m_1 = 4\ m/s^2}$

Problem 2 (30 points)

A bullet of mass m and speed v pierces a wooden pendulum of mass M suspended by a string of length L. The bullet leaves the wooden block with speed $v/3$.



(a) (10 pts.) Please explain why the kinetic energy is <u>not</u> conserved but the linear momentum is conserved during the collision.

During the interaction of the bullet and the block, there is no external force. Thus linear momentum is conserved. The total kinetic energy of the system is NOT conserved, because some of the energy is lost during the interaction. For example, when the bullet hits the block, we hear a sound which is a form of energy. Also, while the bullet pierces the block it changes the formation of block which consumes the kinetic energy of the system.

(b) (10 pts.) Find the speed of block M just after the bullet leaves it.

Total linear momentum is conserved. Thus; $mv + 0 = mv/3 + Mv_M \rightarrow \boxed{v_M = 2mv/3M}$.

(c) (10 pts.) Find the minimum value of v in terms of m, M, L and g such that the bob will swing through a complete circle?.

At the top of the circle: $Mg = Ma_{radial} = Mv_{top}^2/R$. Thus $v_{top}^2 = gR$. Since the mechanical enery is conserved DURING the motion of the block, $Mv_{top}^2/2 + Mg(2R) = Mv_M^2/2 = M(2mv/3M)^2/2$. So, $\boxed{v = \frac{3M}{2m}\sqrt{5gr}}$

Problem 3 (30 points)

A 1.00 kg object is attached to a horizontal spring. The spring is initially stretched by 0.100 m, and the object is released from rest. It moves on a frictionless surface. The next time the speed of the object is zero is 0.500 second later.

Hint: Think about the position function of an object in harmonic motion. Try to find the phase angle using the given conditions, i.e. $x(t=0) = A\cos(\phi) = 0.100\ m$.

(a) (10 pts.) Find the amplitude and the period of the oscillation.

In the system total mechanical energy is conserved, thus $kx^2/2 + mv^2/2$ is conserved. When velocity is zero, the spring is either stretched or compressed to maximum. So for this system maximum compression is $x_m = 0.1\ m$. Recall that for a non damped spring, the amplitude is x_m, i.e. $\boxed{A = 0.1\ m}$.

The next time the speed of the object is zero is 0.500 second later, means that the spring moves from maximum strecthed position to maximum compressed position in 0.5 seconds, which is half of the period T. Thus $\boxed{T = 1\ s}$.

ALTERNATIVE SOL: Initially, $x(t=0) = A\cos(\phi) = 0.100\ m$. Due to energy conservation, $-0.1 \leq x \leq 0.1$. We know that $-1 \leq \cos(\phi) \leq 1$ and amplitude cannot be greater than x_m. So initial condition is only satisfied when $\cos(\phi) = 1$ and $\boxed{A = 0.1\ m}$. Thus $\phi = 0, 2\pi, 4\pi....$

$v(t) = dx(t)/dt = -Aw\sin(wt+\phi) = -0.1w\sin(wt) \rightarrow v(t=0.5s) = -0.1wsin(0.5w) = 0 \rightarrow 0.5w = n\pi \rightarrow w = 2n\pi$. Since $t = 0.5s$ is the first turn, $n = 1$ and $\boxed{w = 2\pi\ /s}$. Using $wT = 2\pi$, $\boxed{T = 1s}$

(b) (10 pts.) Find the spring constant.

$w = 2\pi/s$. Then: $\boxed{w = \sqrt{k/m}} \rightarrow \boxed{k = mw^2 = 4\pi^2\ Nm}$

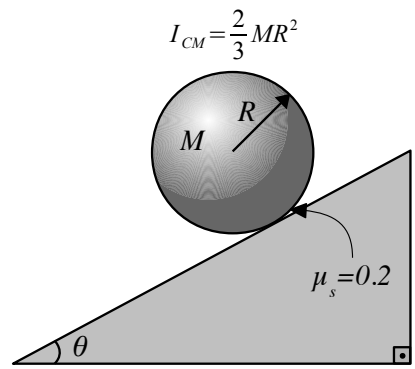
(c) (10 pts.) Find the maximum speed of the object.

Since energy is conserved, $kA^2/2 = mv_m^2/2 \rightarrow \boxed{v_m = A\sqrt{k/m} = Aw = 0.2\pi\ m/s}$.

ALTERNATIVE SOL: We found that $v(t) = -Aw\sin(wt) = -0.2\pi\sin(2\pi\ t)$. Due to sine function, velocity is maximum when $\sin(wt) = -1$. Thus $v_{max} = 0.2\pi\ m/s$

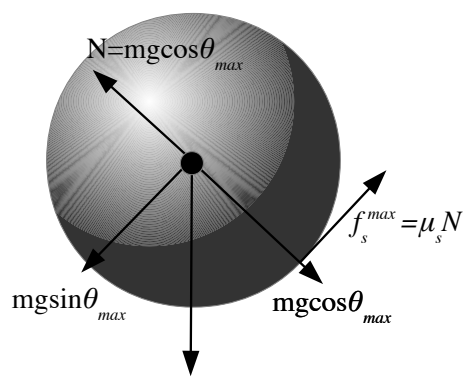
Problem 4 (30 points)

A hollow spherical shell of radius R and mass M rolls downward on an inclined plane with an angle of inclination θ. The coefficient of static friction between the sphere and the surface of incline is $\mu_s = 0.2$. Find the maximum possible value of θ, so that the sphere can still roll without slipping.



Recall that while rolling wirhiur slipping, the relation between angular and linear acceleration (of CM) is $\boxed{a_{CM} = \alpha R}$. Also the friction force due to static friction is $\boxed{F_{\text{static friction}} \leq N\mu_s}$.

The forces acting on the object can be drawn as:



According to free body diagram: $N = Mg\cos(\theta)$. The static friction force is $\boxed{F_{(\text{static friction})} \leq Mg\cos(\theta)\mu_s}$, thus just before slipping, the friction force is at its maximum value, $Mg\cos(\theta_{max})\mu_s$.

Newtons second law states that: $F_{total} = Ma$ and $\tau = I\alpha$ (rotational dynamics).

Hence, $\boxed{Mg\sin(\theta_{max}) - F_{(\text{static friction})} = Ma_{CM}}$

$\boxed{F_{(\text{static friction})} = Mg\sin(\theta_{max}) - Ma_{CM} = Mg\cos(\theta_{max})\mu_s}$(i)

and $\boxed{F_{(\text{static friction})}R = I\alpha = 2MR^2\alpha/3}$(ii).

From (ii), one can get: $F_{(\text{static friction})} = I\alpha = 2Ma_{CM}/3 \rightarrow \boxed{a_{CM} = 3g\cos(\theta_{max})\mu_s/2}$ just before the slipping. Putting this result into (i) one can get:
$Mg\sin(\theta_{max}) - M(3g\cos(\theta_{max})\mu_s/2) = Mg\cos(\theta_{max})\mu_s$

Therefore; $\boxed{\tan(\theta_{max}) = 5\mu_s/2}$

Formula Sheet

DO NOT SEPARATE THIS PAGE!

Scalar Product:

$c = \vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$

$c = AB\cos\phi$

Vector Product:

$\vec{C} = \vec{A} \times \vec{B}$

$C = AB\sin\phi$

$C_x = A_y B_z - A_z B_y$

$C_y = A_z B_x - A_x B_z$

$C_z = A_x B_y - A_y B_x$

Straight-line motion:

$$v_{av-x} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\boldsymbol{\Delta x}}{\boldsymbol{\Delta t}}$$

$$v_x = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta x}}{\boldsymbol{\Delta t}} = \frac{dx}{dt}$$

$$a_{av-x} = \frac{v_{2x} - v_{1x}}{t_2 - t_1} = \frac{\boldsymbol{\Delta v_x}}{\boldsymbol{\Delta t}}$$

$$a_x = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta v_x}}{\boldsymbol{\Delta t}} = \frac{dv_x}{dt}$$

Constant x-acceleration:

$v_x = v_{0x} + a_x t$

$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$

$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$

$$x - x_0 = \left(\frac{v_{0x} + v_x}{2}\right) t$$

Motion in three dimensions:

$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$

$$\vec{v}_{av} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} = \frac{\boldsymbol{\Delta \vec{r}}}{\boldsymbol{\Delta t}}$$

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta \vec{r}}}{\boldsymbol{\Delta t}} = \frac{d\vec{r}}{dt}$$

$$\vec{a}_{av} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \frac{\boldsymbol{\Delta \vec{v}}}{\boldsymbol{\Delta t}}$$

$$\vec{a} = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta \vec{v}}}{\boldsymbol{\Delta t}} = \frac{d\vec{v}}{dt}$$

Circular motion:

$$a_{\text{rad}} = \frac{v^2}{R} \quad a_{\text{rad}} = \frac{4\pi^2 R}{T^2}$$

Relative velocity:

$v_{\text{P/A-}x} = v_{\text{P/B-}x} + v_{\text{B/A-}x}$

$\vec{v}_{\text{P/A}} = \vec{v}_{\text{P/B}} + \vec{v}_{\text{B/A}}$

Newton's 1st Law:

$\vec{R} = \vec{F}_1 + \vec{F}_2 + \ldots = \sum \vec{F}$

$\sum \vec{F} = 0$

Newton's 2nd Law:

$\sum \vec{F} = m\vec{a}$

Newton's 3rd Law:

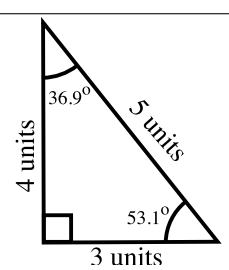
$\vec{F}_{\text{A on B}} = -\vec{F}_{\text{B on A}}$

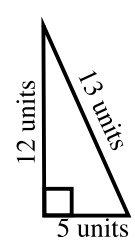
Friction Force:

$f_k = \mu_k n \quad f_s \leq \mu_s n$

$\sin 36.9^\circ = \cos 53.1^\circ = 0.6$

$\sin 53.1^\circ = \cos 36.9^\circ = 0.8$





Formula Sheet

Work done by a force:

$W = \vec{F} \cdot \vec{s} = Fs\cos\phi$

Kinetic energy:

$K = \frac{1}{2}mv^2$

The work-energy theorem:

$W_{\text{tot}} = K_2 - K_1 = \Delta K$

Work done by a varying force:

$W = \int_{x_1}^{x_2} F_x \, dx$

Power:

$P_{\text{av}} = \frac{\Delta W}{\Delta t}$

$P_{\text{av}} = \vec{F} \cdot \vec{v}$

Gravitational and elastic pot. energy:

$W_{\text{grav}} = mgy_1 - mgy_2$

$W_{\text{el}} = \frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2$

Conservation of mech. energy:

$K_1 + U_1 + W_{\text{other}} = K_2 + U_2$

Determining force from potential:

$\vec{F} = -\left(\frac{\partial U}{\partial x}\hat{\imath} + \frac{\partial U}{\partial y}\hat{\jmath} + \frac{\partial U}{\partial z}\hat{k}\right)$

Momentum of a particle:

$\vec{p} = m\vec{v}$

$\sum \vec{F} = \frac{d\vec{p}}{dt}$

Impulse:

$\vec{J} = \sum \vec{F}\Delta t$

$\vec{J} = \vec{p}_2 - \vec{p}_1$

Conservation of momentum:

$\vec{P} = \vec{p}_A + \vec{p}_B + \ldots$

Elastic collision:

$v_{B2x} = \frac{2m_A}{m_A + m_B} v_{A1x}$ (assuming $v_{B1x} = 0$)

Center of mass:

$\vec{r}_{\text{cm}} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + \ldots}{m_1 + m_2 + \ldots}$

Rotational kinematics:

$\omega_z = \frac{d\theta}{dt}$

$\alpha_z = \frac{d\omega_z}{dt}$

$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha_z t^2$

$v = r\omega$

$a_{\tan} = r\alpha$

$a_{\text{rad}} = \frac{v^2}{r} = \omega^2 r$

Torque:

$\vec{\tau} = \vec{r} \times \vec{F}$

Rotational dynamics:

$\sum \tau_z = I\alpha_z$

$K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2$

Rolling without slipping:

$v_{\text{cm}} = R\omega$

Harmonic Motion:

$x(t) = A\cos(\omega t + \phi)$

$\omega = \sqrt{k/m}, \sqrt{g/l}$

$\omega T = 2\pi$

$T = 1/f$